

Joel Oskarsson

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ETH Zurich AI Center, Zurich, Switzerland
Swedish (native) · English (fluent) · German (basic, A2)

Academic Career

2025 – **Postdoctoral Fellow, ETH Zurich AI Center**, Zurich, Switzerland

- Awarded 2-year fellowship in highly competitive call with a **4% acceptance rate**.
- Mentored by Torsten Hoefler (Computer Science), Siddhartha Mishra (Math), and Benedikt Soja (Space Geodesy). Topic: *Modeling the Earth System in Latent Space*.
- Funded by the Swiss AI Initiative, part of effort to develop **AI foundation models**.

2020–2025 **PhD in Computer Science, Linköping University**, 240 ECTS

- Thesis: *Modeling Spatio-Temporal Systems with Graph-based Machine Learning*
- Supervised by Fredrik Lindsten, Per Sidén, Tomas Landelius.
- Affiliated with WASP graduate school.

2015–2020 **Master's in Computer Science and Engineering, Linköping University**, 300 ECTS

- **Exchange year at ETH Zurich**, Zurich, Switzerland (2018–2019).

Research

Summary of publications

See separate document for full list of publications.

- 6 conference papers, 2 journal papers, 7 workshop papers, 3 works under review.
- 424 total citations, h-index 9 (from Google Scholar, November 2026).

Research and Career Grants

- Co-applicant for ICAIN-IGAIP project grant (320,000 CHF ≈ 345,000 EUR, **4/40 applications funded**, 2026).
  Project *AI-Powered Early Warning and Profiling for Displaced Populations*, with Christina Humer (PI, ETHZ) and the International Organization for Migration.
- Research Visit Scholarship (60,000 SEK ≈ 5,200 EUR), WASP Program.
- Co-author of applications for compute resources from NSC, Sweden and CSCS, Switzerland.

Software Research Output

Creator and maintainer of the *Neural-LAM* **open source library** for AI weather forecasting research.

- Translates research findings into direct community impact through open-source tooling.
- **Over 300 GitHub forks and stars**, reflecting widespread usage.
- Used by multiple **international research groups**, basis for many peer-reviewed publications.
- Organizational participant in Google Summer of Code 2026, personally acting as project mentor.

Invited Talks

2026 KCDS Summer School on Graph Neural Networks, Karlsruhe Institute of Technology (KIT)
GNNs for Spatio-Temporal and Earth System Modeling

2025 **Google DeepMind**
Probabilistic Weather Forecasting with Hierarchical Graph Neural Networks

2024 RISE Learning Machines Seminars
Frontiers in Machine Learning for Weather Forecasting

European Centre for Medium-Range Weather Forecasts (ECMWF)
Graph-based Machine Learning for Weather

University of New South Wales (UNSW)
Graph-based Machine Learning for Spatio-Temporal Data, with Application to Traffic and Weather Forecasting

Selected further invited talks: WeatherGenerator Dissemination Workshop (2026); MeteoSwiss (2024); ETH Zurich (2024); Uppsala University (2024); Danish Meteorological Institute (2023).

Research Visits

2023 Five-week research visit to **University College London (UCL)**, London, U.K.
Hosted by Prof. Marc Deisenroth, Sustainability and Machine Learning Group.

Shorter visits: Norwegian Meteorological Institute, Oslo (2026); ECMWF, Bonn (2024); ETH Zurich + MeteoSwiss, Zurich (2024); Finnish Center for AI + Aalto University, Helsinki (2022, WASP International Study Visit).

Presentations at Scientific Events

- Presentations at top **machine learning conferences**, and accompanying workshops: NeurIPS (2024: poster; 2023: workshop poster), ICML (2024: workshop poster), AISTATS (2023: poster).
- Selected for **oral presentation** at the Machine Learning for Earth System Modelling Workshop, Bonn, Germany, in **three consecutive years** (2023, 2024, 2025).
- Oral presentation at the KVA Symposium on AI for Science, Stockholm (2025)
- Oral presentation, poster and panel discussion at the ESA-ECMWF Workshop on Machine Learning for Earth Observation, Frascati, Italy (2024)

Selected further presentations: Climate Informatics, Lausanne, Switzerland (2026, oral); Learning on Graphs and Geometry Meetup, Uppsala (2025, oral); NORDSTAT, Gothenburg (2023, oral); Swedish Symposium on Image Analysis and Deep Learning, Norrköping (2023, poster);

Community and Academic Service

Reviewing and Editorial Service

Area Chair: ICLR (2027).

Reviewer: NeurIPS (2024; 2026: *Top Reviewer*); ICLR (2025, 2026); AISTATS (2023, 2024, 2026); Climate Informatics (2026); Geoscientific Model Development (2026); Quarterly Journal of the Royal Meteorological Society (2026); Geophysical Research Letters (2025); SOLA (2025); Geomatics, Natural Hazards and Risk (2025, 2026); CCAI workshop (2024, 2025).

Organizing and Other Service

2026 Co-organizer of the *AI + Environment Summit 2026*, Zurich. Responsible for session on AI for weather and climate, and poster session.

2025–2026 Reviewer for applications to ETH Zurich AI Center Fellowship 2026.

2023–2025 Co-organizer of monthly online meetups for ML limited area weather forecasting researchers.

2022–2025 Organizer of the *STIMA ML Reading Group*, Linköping University.

Teaching, Supervision and Mentoring

- Supervisor of **10 master's theses** (8 LiU and 2 ETHZ, 1 ongoing) and 1 bachelor's thesis (ETHZ, ongoing).
- Mentored junior PhD students through collaborative research, co-authorship, and project guidance.
- Co- and guest-lecturer for courses *AI Center Projects in Machine Learning Research* (ETHZ) and *Time Series and Sequence Learning* (LiU).
- **Co-course-developer** of the *Foundations of Machine Learning* online course (LiU, 2021–2024).
- Pedagogical training: *Becoming a Teacher in Higher Education*, 6 ECTS (LiU, 2021).

Other teaching assistant work (2016–2024): Advanced Machine Learning (5×), Computational Statistics, Foundations of ML, Building AI, Introduction to Python, etc. All at LiU.

Media Coverage and Scientific Outreach

- *De vill göra väderprognoser mer pålitliga – med hjälp av AI*, SVT (2024): **Television news feature**.
- *Förbättrade väderprognoser med AI*, Linköping University (2023): News article on University website.
- *Forskningsframsteg inom Maskininlärning för Väderprognoser* (2025): **Popular science article** in Polarfront, journal of the Swedish meteorological society.
- *AI för prognosmeteorologer* (2025): Lecture for forecast meteorologists about AI weather forecasting.